Lesson 14: Transformations

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2026-11-16

Learning Objectives

1. Use the simulation method (simulate, transform, summarize) to transform a random variable
2. Use simulation to approximate probabilities and the shape of the distribution of a *linear* transformation
3. Use simulation to approximate probabilities and the shape of the distribution of a *nonlinear* transformation

Distributions of transformations of random variables

- Often make transformations of RVs
  - Unit conversions (Celsius to Fahrenheit, minutes to hours)
  - Log-transforming a skewed biomarker
  - Turning a waiting time into a rate
- A function of a **random variable** is a random variable
  - If X is a random variable and g is a function then $Y = g(X)$ is a random variable
  - Since $g(X)$ is a random variable it has a **distribution**
- Two types:
  - Linear rescalings: $g(u) = a + bu$
  - Nonlinear transformations: e.g. $g(u) = u^2$, $g(u) = \log(u)$
- Our main tool today: **simulate** X, transform every simulated value, then summarize

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Our main tool: simulate, transform, summarize

The simulation method for transformations

To learn about the distribution of $Y = g(X)$:

1. **Possible values:** Find the possible values of Y (graph g over the possible values of X)
2. **Simulate:** Simulate many values of X
3. **Transform:** Apply g to every simulated value: `mutate(y = g(x))`
4. **Summarize:**
   - Probabilities: `mean(y <= c)` approximates $P(Y \leq c)$
   - Shape: a histogram of `y`
   - Center and spread: `mean(y)`, `sd(y)`

- This works for **any** function g, as long as we can simulate X
- We do not need the CDF of X, the CDF of Y, or a `p*()` function for Y

For step 1: possible values and rewriting events

- For linear *and* nonlinear transformations, rewriting an event about $Y = g(X)$ as an event about X helps us:
  - Find the possible values of Y (Step 1 in our main tool)
  - Check that our simulation summary is counting the right thing (Step 4)
  - Find an exact answer when we *can* find probabilities for X (a `p*()` function or a rectangle's area)
- Examples:

$Y = g(X)$	**Event about Y**	**Same event about X**
$Y = 1 - X$ (linear)	$Y \leq -0.5$	$X \geq 1.5$ (multiplying by -1 flips the inequality!)
$Y = \log(X)$ (nonlinear)	$Y \leq 0.5$	$X \leq e^{0.5}$
$Y = X^2$ (nonlinear)	$Y \leq 0.25$	$-0.5 \leq X \leq 0.5$ (two sides!)
$Y = \frac{1}{X}$ (nonlinear)	$Y > 2$	$0 < X < 0.5$ (the inequality flips, and X must be positive!)

- Graph g to see which values of X land in the event

For step 2 simulation: four ways

How do we get simulated values of X in step 2?

1. **Named distribution:** use its `r*()` function from Lessons 12 and 13 (`runif()`, `rexp()`, `rgamma()`, `rnorm()`)

2. **Build it from simpler pieces:** like the Gamma as a sum of exponentials (Lesson 13)

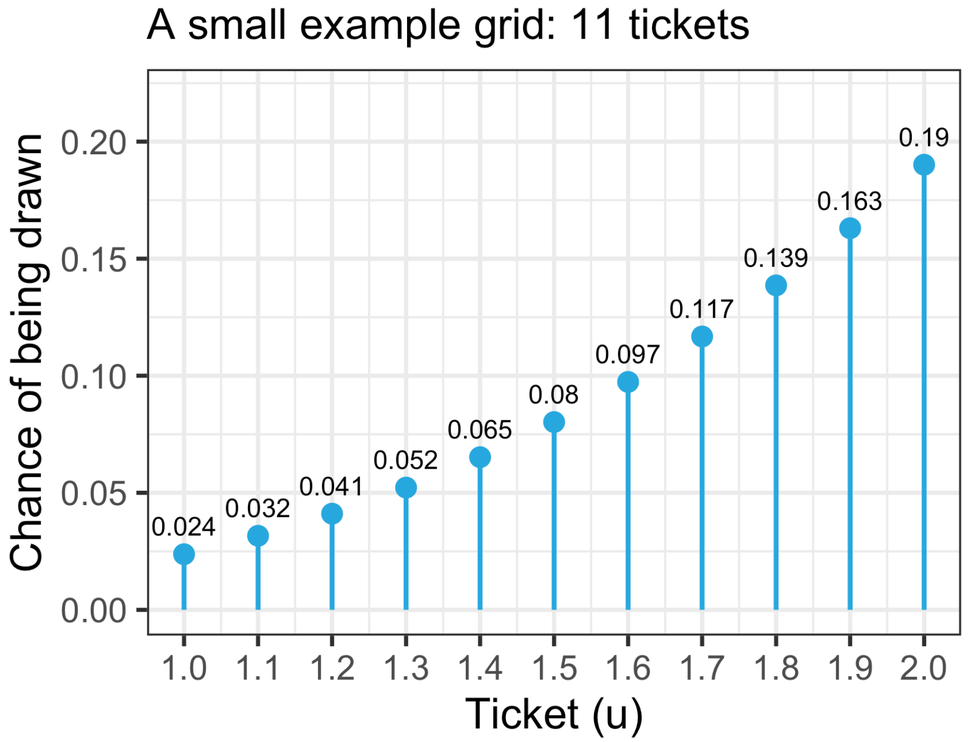
3. **Have the CDF?** Use the **inverse CDF method** from Lesson 9
   - Plug Uniform(0, 1) values into F_X^{-1}
   - Needs a CDF we can write down *and* solve for x

4. **Only have the pdf?** Use **grid sampling** (next slide)
   - Only needs the *height* of the pdf: no CDF required

Grid sampling: the steps

- Let U have pdf $f_U(u) = \frac{4}{15}u^3$ for $1 \leq u \leq 2$. There is no `r*()` function for U!
- Idea: a box model (Lesson 2), where tickets in taller parts of the pdf are more likely to be drawn

1. **Find the possible values** of U: from 1 to 2
2. **Make a grid:** many evenly spaced values across the possible values. Each value is a ticket in the box
   - e.g. 1, 1.0001, 1.0002, ..., 2 (10,001 tickets)
3. **Find the pdf's height** at each ticket: $f_U(1) \approx 0.27$, ..., $f_U(2) \approx 2.13$
4. **Draw many tickets with replacement.** Each draw is one simulated value of U
   - Each ticket has a chance of being drawn proportional to its height
5. **Check:** a histogram of the draws should follow the pdf



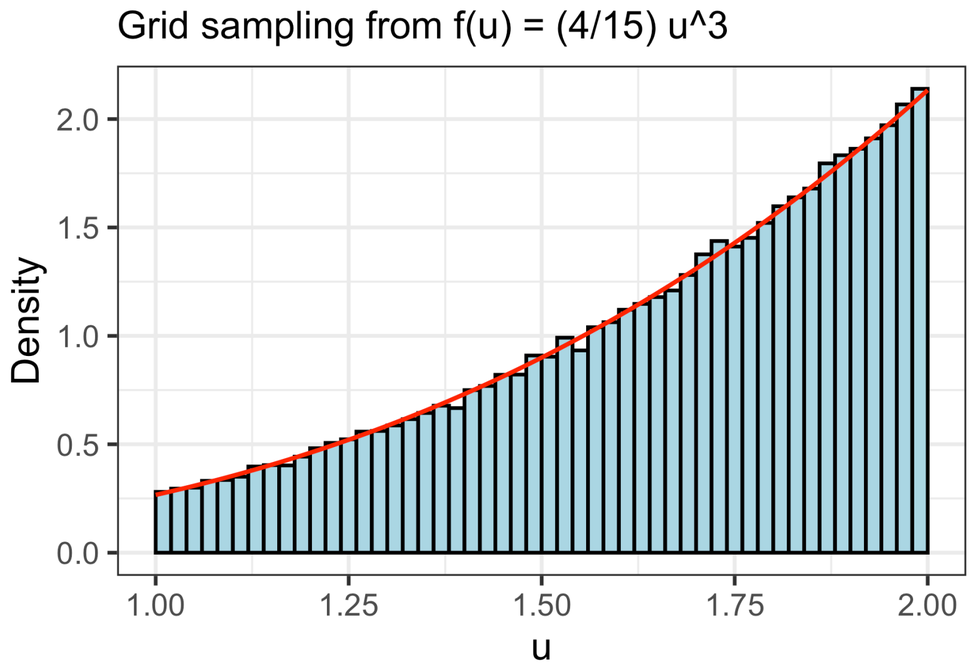
Real grid has 10,001 tickets, so draws look continuous

Grid sampling: code and plot

- Simulate 100,000 values of U, with $f_U(u) = \frac{4}{15}u^3$ for $1 \leq u \leq 2$, by grid sampling

```
1  set.seed(1111)                                          ①
2  reps <- 100000                                          ②
3
4  u_grid <- seq(
5    from = 1,
6    to = 2,
7    length.out = 10001
8    )                                                     ③
9
10 u_sims <- tibble(sim = 1:reps) |>                       ④
11   mutate(
12     u = sample(u_grid,                                  ⑤
13                size = n(),
14                replace = TRUE,
15                prob = 4/15 * u_grid^3                  ⑥
16                )
17   )
```

▶ Plot code



The larger we make `reps` and `length.out`, the closer we get to the exact pdf.

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Linear rescaling

Definition: Linear Rescaling

A **linear rescaling** is a transformation of the form $g(u) = a + bu$, where a and b are constants

- Thus, if we have a random variable, X, then a linear rescaling of X could be $M = g(X) = a + bX$
- For example, converting temperature from Celsius to Fahrenheit using $g(u) = 32 + 1.8u$ is a linear rescaling.

Example 1: Linear rescaling of U

Let U be a random variable with $f_U(u) = \frac{4}{15}u^3$ for $1 \leq u \leq 2$. Define $V = 1 - U$

1. What are the possible values of V?

2. Is V the same random variable as U?

3. Rewrite the event $V \leq -0.5$ in terms of U. Then approximate $P(V \leq -0.5)$ with simulation.

Example 1: Linear rescaling of U

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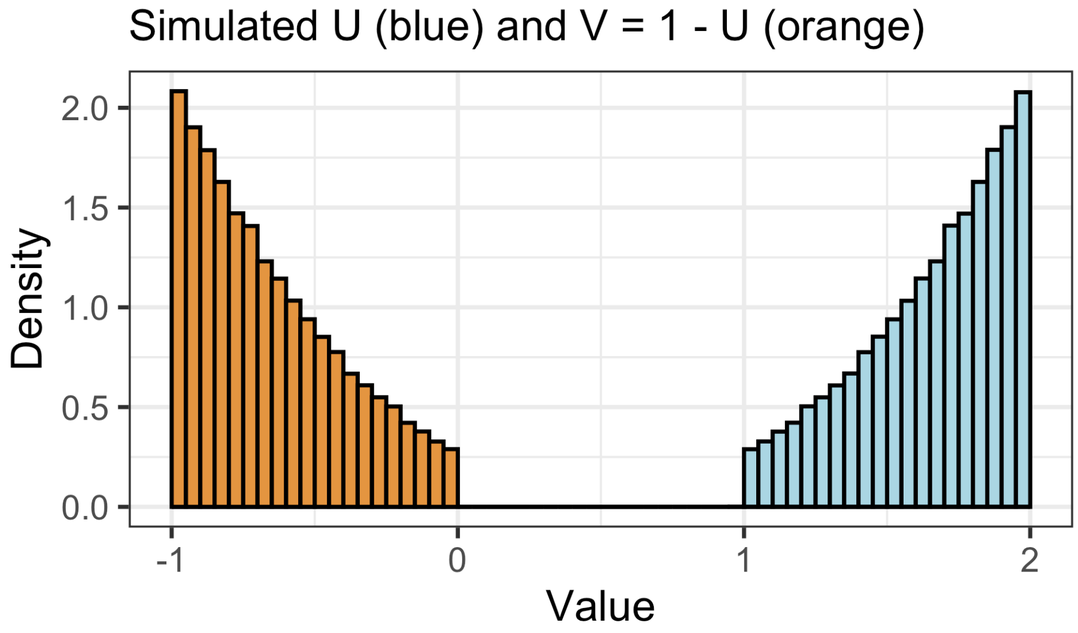
3. Rewrite the event $V \leq -0.5$ in terms of U. Then approximate $P(V \leq -0.5)$ with simulation.

```
1  u_sims <- u_sims |>                                  ①
2    mutate(v = 1 - u)                                  ②
3
4  u_sims |>                                            ③
5    summarise(
6      prob_V = mean(v <= -0.5),                        ④
7      prob_U = mean(u >= 1.5)                          ⑤
8    )
```

```
# A tibble: 1 × 2
  prob_V prob_U
   <dbl>  <dbl>
1  0.731  0.731
```

For reference, the exact answer is $P(V \leq -0.5)$ ≈ 0.7292

▶ Plot code



What happens to the distribution?

- $V = 1 - U$ flips U around: same shape, mirrored

- In general, linear rescaling can **shift** (add a), **stretch or condense** (multiply by $|b|$), and **flip** ($b < 0$) a distribution
  - Stretching the values out by $|b|$ makes the histogram $|b|$ times wider…
  - …so it must get $|b|$ times shorter to keep the total area equal to 1

- We say this is a preservation of shape: only scaled

- Center and spread change the way we saw in Lessons 10 and 11:
  - $\mathbb{E}[a + bX] = a + b\,\mathbb{E}[X]$
  - $Var(a + bX) = b^2\,Var(X)$ and $SD(a + bX) = |b|\,SD(X)$

Linear rescaling of named distributions

For most named distributions, a linear rescaling stays in the **same family**, with new parameters:

If $X \sim$	then	is
$\text{Normal}(\mu, \sigma^2)$	$a + bX$	$\text{Normal}(a + b\mu,\ b^2\sigma^2)$
$\text{Uniform}[l, r]$	$a + bX$ (with $b > 0$)	$\text{Uniform}[a + bl,\ a + br]$
$\text{Exponential}(\lambda)$	cX (with $c > 0$)	$\text{Exponential}(\lambda/c)$
$\text{Gamma}(r, \lambda)$	cX (with $c > 0$)	$\text{Gamma}(r,\ \lambda/c)$

- Exponential and Gamma: only multiply, do not shift! $X + 3$ can't be close to 0, so it is not Exponential
- **Standardizing** is a linear rescaling: if $X \sim \text{Normal}(\mu, \sigma^2)$, then

$$Z = \frac{X - \mu}{\sigma} \sim \text{Normal}(0, 1)$$

Linear rescaling: final words

- Remember, do NOT confuse a **random variable** with its **distribution**
  - The **random variable** is the numerical quantity being measured
  - The **distribution** is the long run pattern of variation of many observed values of the random variable

- Ultimately, a transformed **random variable** does NOT have the same **distribution** - *possible values change*

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Nonlinear transformations

- What happens when we make a **nonlinear transformation**, like a logarithmic or square root transformation?

- Nonlinear transformations do *not* necessarily preserve the distribution shape

- Examples of nonlinear transformations:
  - $g(u) = u^2$
  - $g(u) = \sqrt{u}$
  - $g(u) = \log(u)$
  - $g(u) = e^u$
  - $g(u) = \dfrac{1}{u}$

Example 2: Log transformation of U

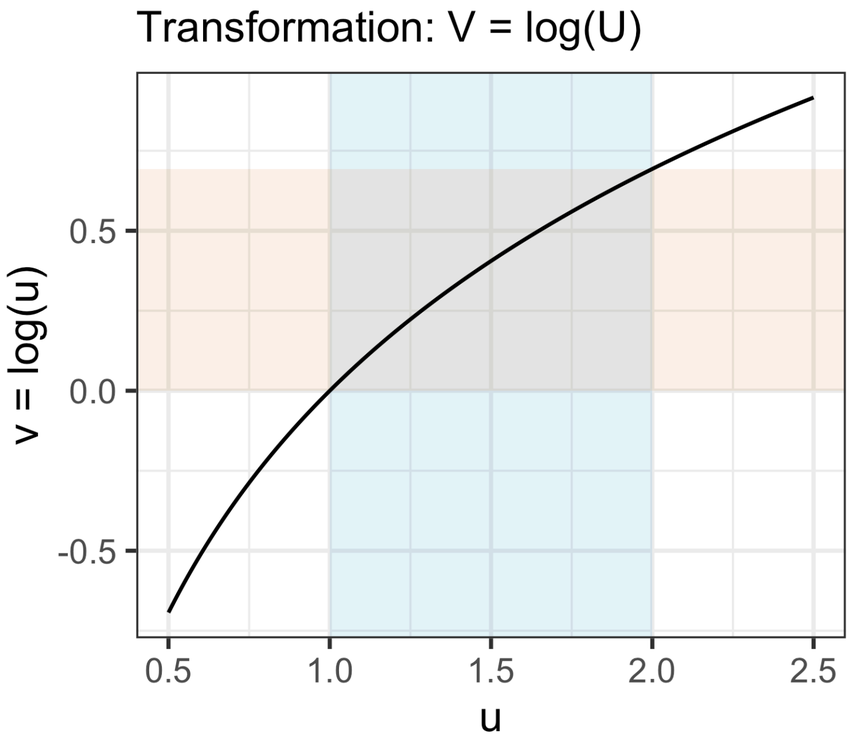
Let U be a random variable with $f_U(u) = \frac{4}{15}u^3$ for $1 \leq u \leq 2$. Define $V = \log(U)$

1. What are the possible values of V?
2. Rewrite the event $V \leq 0.5$ in terms of U.
3. Approximate $P(V \leq 0.5)$ with simulation.

Example 2: Log transformation of U

Let U be a random variable with $f_U(u) = \frac{4}{15}u^3$ for $1 \leq u \leq 2$. Define $V = \log(U)$

1. What are the possible values of V?



Example 2: Log transformation of U

Let U be a random variable with $f_U(u) = \frac{4}{15}u^3$ for $1 \leq u \leq 2$. Define $V = \log(U)$

2. Rewrite the event $V \leq 0.5$ in terms of U.

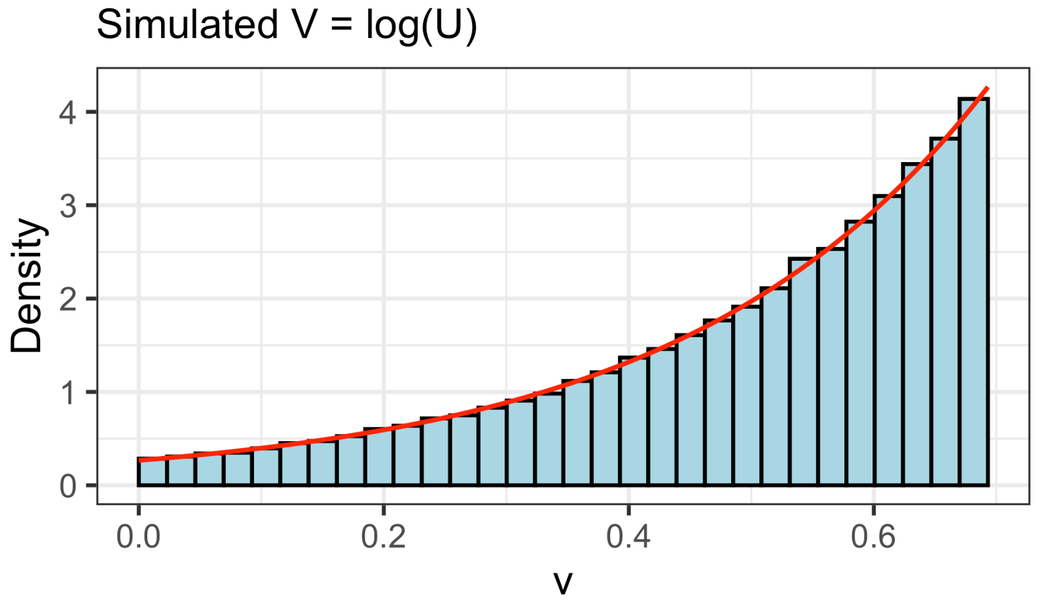
3. Approximate $P(V \leq 0.5)$ with simulation.

```
1 u_sims <- u_sims |>                                  ①
2   mutate(log_u = log(u))                             ②
3
4 u_sims |>                                            ③
5   summarise(
6     prob_V = mean(log_u <= 0.5),                     ④
7     prob_U = mean(u <= exp(0.5))                     ⑤
8   )
```

```
# A tibble: 1 × 2
  prob_V prob_U
   <dbl>  <dbl>
1  0.423  0.423
```

For reference, the exact answer is $P(V \leq 0.5)$
≈ 0.4260

▶ Plot code



Example 3: Squaring a Uniform RV

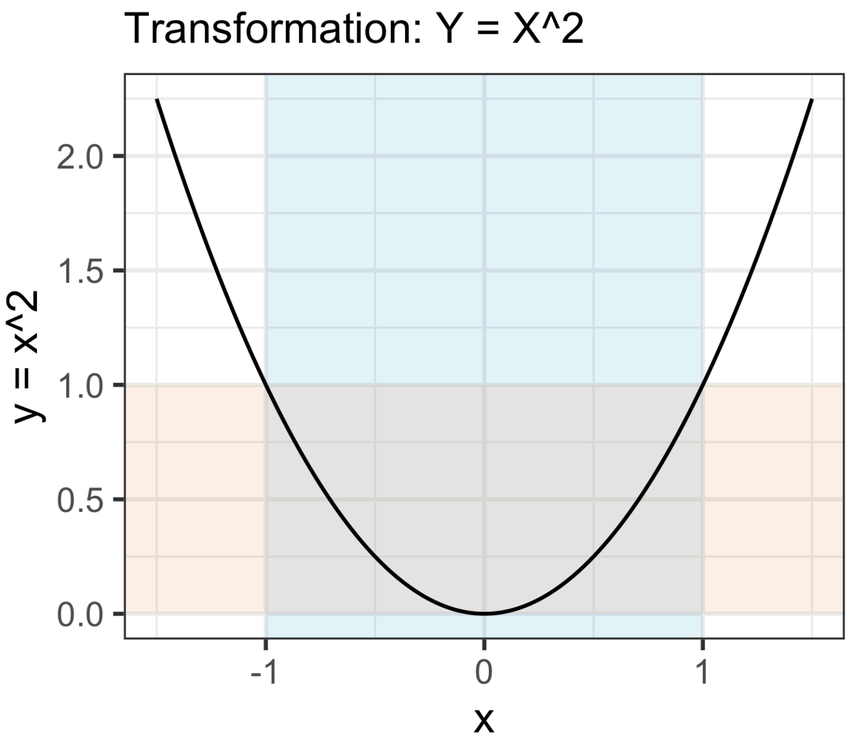
Let $X \sim \text{Uniform}[-1, 1]$, so $f_X(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$. Define $Y = X^2$

1. What are the possible values of Y?
2. Rewrite the event $Y \leq 0.25$ in terms of X.
3. Find $P(Y \leq 0.25)$ exactly using the distribution of X. Check it with simulation.

Example 3: Squaring a Uniform RV

Let $X \sim \text{Uniform}[-1, 1]$, so $f_X(x) = \frac{1}{2}$ for $-1 \leq x \leq 1$. Define $Y = X^2$

1. What are the possible values of Y?



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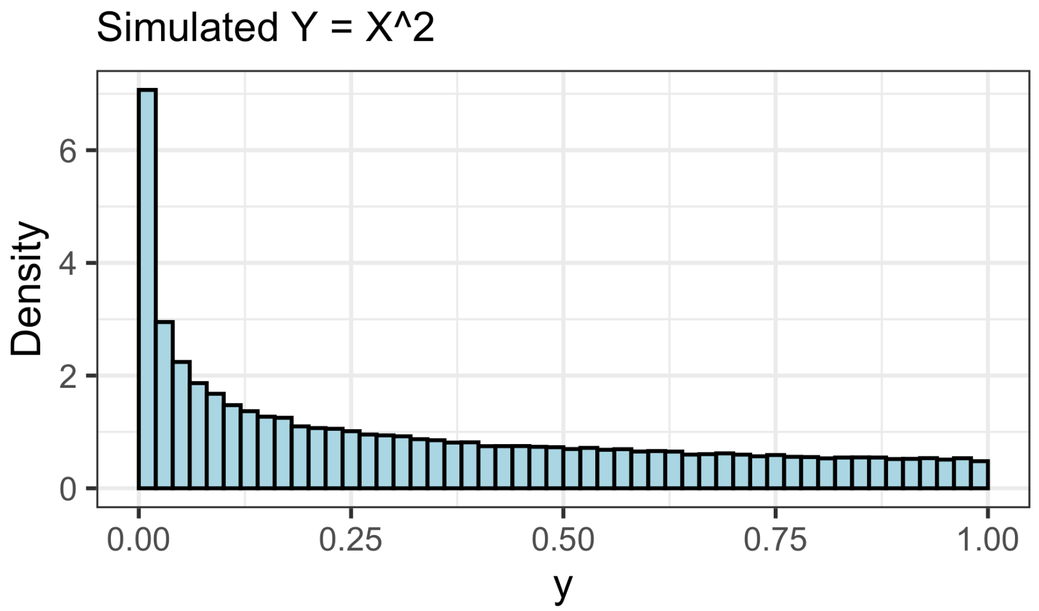
3. Find $P(Y \leq 0.25)$ exactly using the distribution of X.

3. Check $P(Y \leq 0.25)$ with simulation.

```
 1 x_sims <- tibble(sim = 1:reps) |>                    ①
 2   mutate(
 3     x = runif(n(), min = -1, max = 1),               ②
 4     y = x^2                                          ③
 5   )
 6
 7 x_sims |>                                            ④
 8   summarise(
 9     prob_Y = mean(y <= 0.25)
10   )
11
12 punif(0.5, min = -1, max = 1) -                      ⑤
13   punif(-0.5, min = -1, max = 1)
```

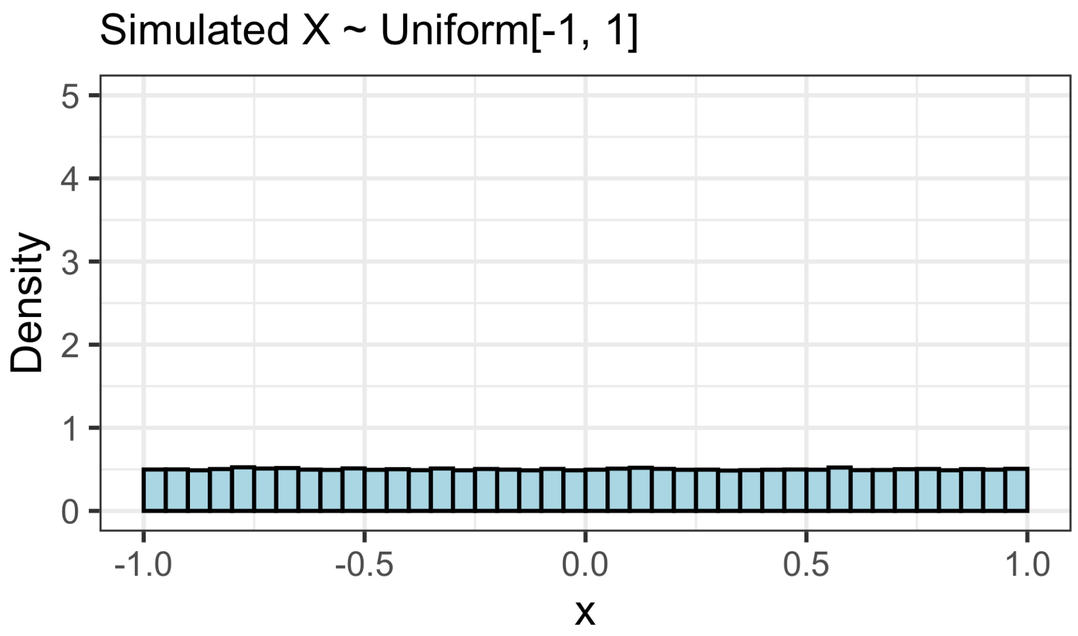
```
# A tibble: 1 × 1
  prob_Y
   <dbl>
1  0.498
[1] 0.5
```

▶ Plot code



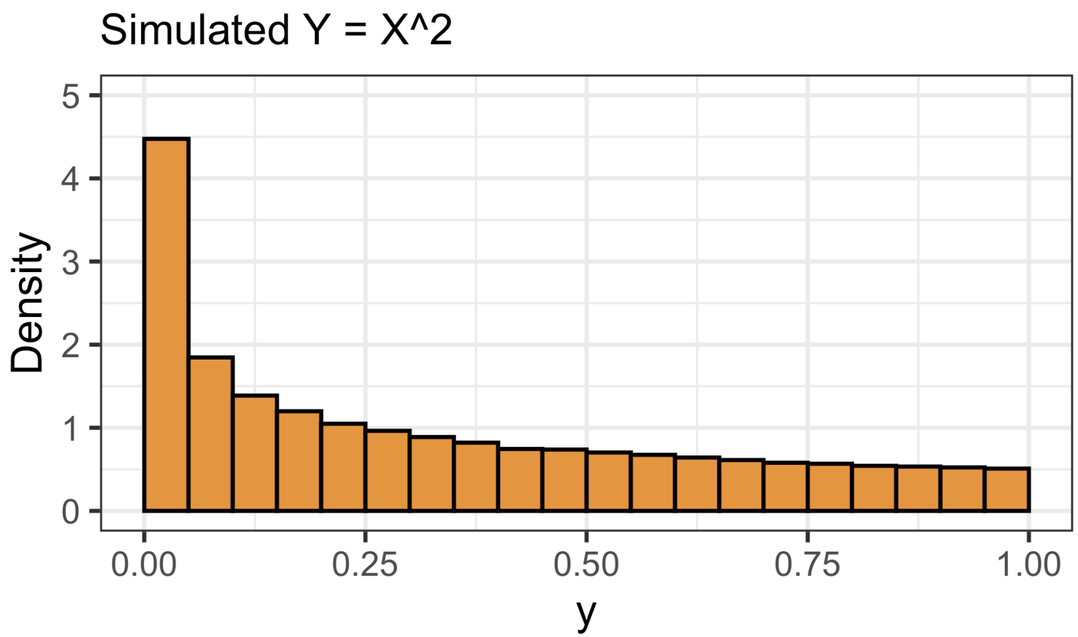
How does the distribution of $Y = X^2$ compare to the distribution of X?

▶ Plot code



- X is flat: every part of $[-1, 1]$ is equally likely

▶ Plot code



- $Y = X^2$ piles up near 0: the shape changed!
  - Half of the values of X (from -0.5 to 0.5) get squeezed into $[0, 0.25]$

How accurate is a simulation? (1/2)

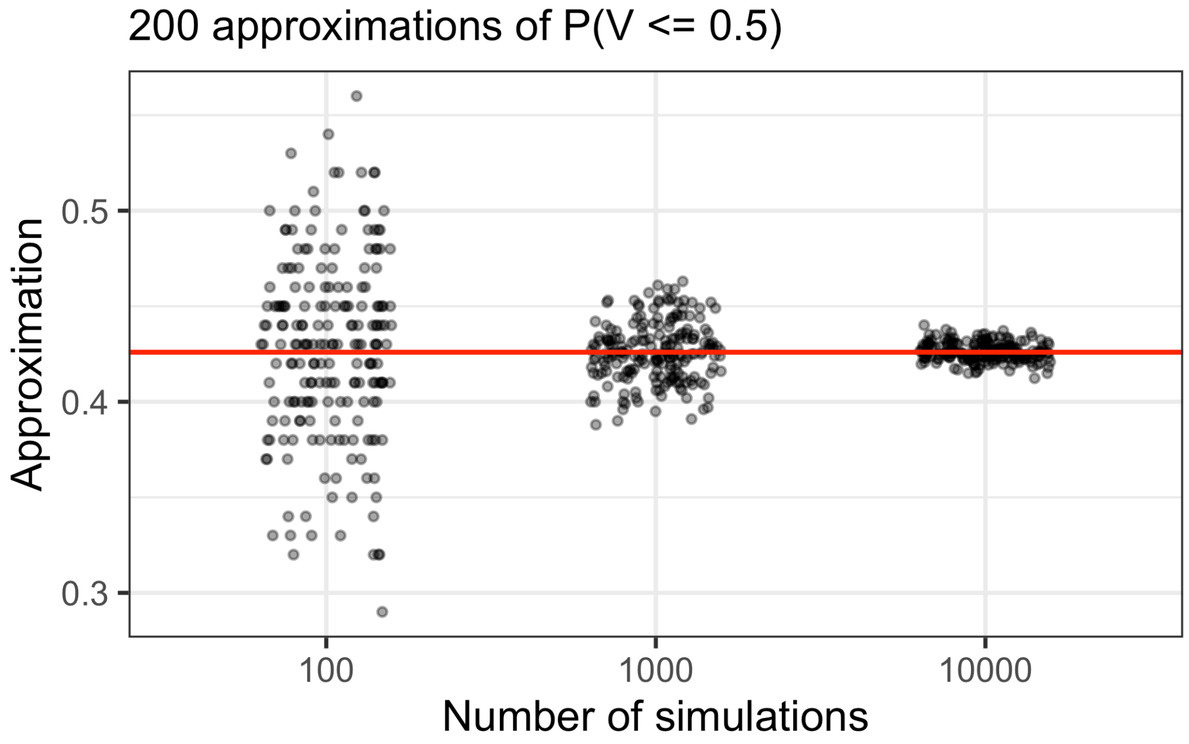
- A single simulation gives an **approximation**. Run it again with a new seed and the answer moves a little
- How much it moves depends on the number of simulations
- Let's repeat the simulation from Example 2 **200 times** for each of 100, 1,000, and 10,000 simulations

▶ Measure average estimate and spread

```
# A tibble: 3 × 3
  n_sims avg_estimate  spread
   <dbl>        <dbl>   <dbl>
1    100        0.428 0.0480
2   1000        0.426 0.0157
3  10000        0.426 0.00491
```

- 100 times more simulations makes the spread about 10 times smaller
  - We will see this again with the Central Limit Theorem

▶ Plot code



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Summary of transformations

Summary of changes for linear and nonlinear transformations:

	Linear rescaling	**Nonlinear transformation**
Changes the distribution?	Yes	Yes
Changes the "shape"?	No	Usually
How to find probabilities	Same family with new parameters, or rewrite the event in terms of X	Simulate, transform, summarize; rewrite the event in terms of X to check

- The **simulation method** works for any transformation: simulate X, transform, summarize
- Rewrite events about Y in terms of X to find possible values, check your simulation, and get exact answers when you can

Some more reference slides

Example 4: Z^2 and the chi-squared distribution

Let $Z \sim \text{Normal}(0, 1)$ and $W = Z^2$.

1. What are the possible values of W?
2. Rewrite $W \leq 4$ in terms of Z, and find $P(W \leq 4)$ with `pnorm()`.
3. In Lesson 13, we said $Z^2 \sim \chi^2_{(1)}$. Check with `pchisq()` and with simulation.

Example 4: Z^2 and the chi-squared distribution

Let $Z \sim \text{Normal}(0, 1)$ and $W = Z^2$.

1. What are the possible values of W?

2. Rewrite $W \leq 4$ in terms of Z, and find $P(W \leq 4)$ with `pnorm()`.

3. In Lesson 13, we said $Z^2 \sim \chi^2_{(1)}$. Check with `pchisq()` and with simulation.

```
 1  z_sims <- tibble(sim = 1:reps) |>                    ①
 2    mutate(
 3      z = rnorm(n(), mean = 0, sd = 1),
 4      w = z^2                                          ②
 5    )
 6
 7  z_sims |> summarise(prob_W = mean(w <= 4))           ③
 8
 9  pnorm(2) - pnorm(-2)                                 ④
10  pchisq(4, df = 1)                                    ⑤
```

```
# A tibble: 1 × 1
  prob_W
   <dbl>
1  0.954
[1] 0.9544997
[1] 0.9544997
```

▶ Plot code

